

SmartStudy AI: Academic Planning and Productivity Assistant

Executive Summary

SmartStudy AI is an academic productivity tool designed to help college students manage assignments, deadlines, and study schedules more effectively. Many students struggle to organize their workload, often leading to stress, procrastination, and missed deadlines. SmartStudy AI uses multiple AI agents that work together to break assignments into smaller tasks, create study schedules, and adjust plans when students fall behind. The goal of this project is to demonstrate how agentic AI can improve time management and productivity while providing a simple and helpful user experience.

Problem Statement and Motivation

Many college students have difficulty managing multiple classes, assignments, and responsibilities at the same time. Students often underestimate how long assignments will take or forget important deadlines. Traditional tools like planners and calendar apps require students to manually organize their schedules, which can be time-consuming and sometimes ineffective. This problem is important because productivity and time management tools are in high demand in education. Universities and students are constantly looking for ways to improve academic performance and reduce stress. SmartStudy AI addresses this issue by automatically organizing assignments and helping students plan their study time. The main target users for this system are college and university students, especially those taking multiple courses or managing busy schedules. An agentic AI solution is appropriate because assignment planning involves several connected steps, including breaking down tasks, scheduling study time, tracking progress, and adjusting plans when changes occur. Multiple AI agents can work together to complete these tasks faster and more efficiently than traditional planning tools or manual scheduling.

Use Cases and User Scenarios

One common scenario is when a student wants to create a weekly study plan. The student enters assignment details, such as due dates and difficulty level. The planner agent breaks assignments into smaller tasks. The scheduler agent then organizes those tasks into a study schedule based on available study time. The reviewer agent checks if the schedule is balanced and realistic before showing the final plan to the student. Another scenario occurs when a student falls behind on work. If a student misses a task or deadline, the catch-up agent reorganizes the schedule and suggests ways to complete remaining assignments on time. The reviewer agent then checks the updated schedule to make sure it is still manageable. A third scenario involves tracking progress. The system records completed tasks and analyzes study habits. The feedback agent provides suggestions to improve productivity and help students stay organized.

Initial System Design

The system will include several agents with specific roles. The user interface agent will collect assignment information and display schedules and progress updates. The planner agent will break assignments into smaller tasks. The scheduler agent will organize tasks into study schedules based on deadlines and available time. The catch-up agent will adjust schedules if students fall behind. The reviewer agent will check schedules to ensure they are realistic and balanced. The project will use Opal as the main platform to build and manage the multi-agent workflow. Opal will allow the agents to communicate, process user input, and generate study plans through an easy to use interface. The system will also use the OpenAI API to generate task breakdowns, schedules, and productivity feedback. Optional integration with Google tools may be included to improve usability.

Feasibility and Risks

This project is technically feasible because it uses existing AI tools and simple web development frameworks. The system will mainly rely on text-based inputs, which reduces technical difficulty. Some challenges may include AI misunderstanding assignment details or producing unrealistic schedules. To reduce this risk, users will be able to review and edit AI-generated results. Another risk involves API limitations or system errors. Backup manual input options will be available if tools fail. Privacy is also important because student data will be stored. The system will store data securely and only use necessary information. Project constraints include limited development time, API usage limits, and available computing resources. However, the project design focuses on simplicity to ensure completion within the course timeline.

Project Timeline

During the first two weeks, the proposal and system design will be finalized. Weeks three and four will focus on building the front-end interface. Weeks five and six will involve developing the multi-agent workflow and connecting agents. Weeks seven and eight will include testing and collecting user feedback. The final weeks will focus on system improvements, documentation, and presentation preparation.

Conclusion

SmartStudy AI aims to solve common time management challenges faced by students by providing automated assignment planning and productivity support. The multi-agent AI system allows specialized agents to collaborate and create personalized study plans. This project demonstrates how agentic AI can improve academic organization while remaining practical and achievable within the course timeline.